Akshat Kankani

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Education

Manipal Institute of Technology Bengaluru, KA
B.Tech in Computer Science (Artificial Intelligence) — CGPA: 8.11 *July 2023 – May 2027*

Coursework: *Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Machine Learning, Software Engineering*

Projects

Strata | *C++20, POSIX APIs (file I/O, fsync, signals), Multithreading, Make, GitHub Actions (CI)* GitHub

- Built a **C++ time-series storage engine** with **write-ahead logging**, crash recovery and LSM-style compaction.
- Implemented **Gorilla-based compression** using delta-of-delta and XOR encoding, reducing storage by **4.28×**.
- Built an **inverted index** for high-cardinality queries with **sub-microsecond p50 latency** across 1M series.
- Proved crash safety via **deterministic fault injection**, killing the process at 5 exact code points mid-write/compaction with **zero data loss**.

Low-Latency Matching Engine | *C++20, CMake, GoogleTest, GitHub Actions (CI)* GitHub

- Built a **C++ limit-order-book matching engine** with **price-time priority**, partial fills, and O(1) order cancellation.
- Tested four data-structure designs and improved throughput to **14.5M orders/sec**, a **3.9× improvement**.
- Profiled hardware performance counters to identify a branch-misprediction bottleneck, reducing its cost by half and **improving throughput by over 100%.**
- Added **differential testing** and CI with **71 tests** and **99% line coverage**.

Relay | *Python, FastAPI, SQLAlchemy, PostgreSQL/pgvector, Redis, RAG, GitHub Actions (CI)* Live Demo | GitHub

- Built a **context engine** that connects **GitHub, Slack, and Jira** into 8 cross-system query modes to help teams find code and context, investigate issues and experts, and capture and share engineering knowledge.
- Implemented **hybrid keyword/vector retrieval** and **relevance scoring**, separating true matches from noisy correlations.
- Added **asynchronous LLM processing** with **Celery/Redis** and integrated **four LLM providers**, backed by **300+ tests**.

Technical Skills

Languages: C++, Python, SQL
Databases: PostgreSQL, MongoDB, Redis
Developer Tools: Git, GitHub, Linux, Postman, Pytest, GitHub Actions
Backend & Infrastructure: FastAPI, REST APIs, Docker, Kubernetes, AWS(EC2, S3), Celery, SQLAlchemy, CI/CD
Systems: System Design, Concurrency & Multithreading, Distributed Systems, Performance Optimization
AI & ML: NumPy, Pandas, scikit-learn, PyTorch, LLMs, RAG, MCP

Achievements & Certifications

- Solved **300+** Data Structures & Algorithms problems in **C++** across LeetCode and other coding platforms.
- Achieved a peak **LeetCode Contest Rating of 1862** (**Knight**) through consistent participation in weekly and biweekly coding contests.
- Introduction to Model Context Protocol (MCP) Certification — [Anthropic]